HACKATHON 2025 • BIS STANDARDS TRACK

ARIVANTI

BIS Standards Recommendation Engine

THE PROBLEM

Indian MSEs spend **weeks** identifying which BIS regulations apply to their products — delaying certification, increasing costs, and stalling market entry.

450+ IS Standards

Weeks avg. compliance time

MSEs affected across India

Turning product descriptions into precise BIS standards — in seconds.

02 SOLUTION OVERVIEW

ARIVANTI converts a product description into ranked BIS standards with rationale — instantly.

Hybrid Retrieval

FAISS dense vectors + BM25 sparse search, fused for maximum precision on both semantic and keyword queries.

Intent Graph

Deterministic domain routing using weighted keyword graphs — ensures search stays within the relevant IS category.

Zero Hallucinations

LLM is strictly constrained to retrieved context. Metadata pre-filtering eliminates cross-domain noise before generation.

Self-Healing System

Timeout-aware fallback: if local LLM fails, structured results from FAISS are served directly — zero downtime.

Input: Product description → Output: Top 3–5 BIS Standards + rationale in plain English

03 SYSTEM ARCHITECTURE

4-Stage Hybrid RAG Pipeline

01	02	03	04
Ingest & Index	**Intent Graph**	**O(1) Filter + FAISS**	**LLM Generation**
PDF → JSON chunks all-MiniLM-L6-v2 FAISS + metadata.json	Domain detection IS code lookup 23 BIS sections	Metadata pre-filter Cosine similarity Top-K retrieval	Llama 3 (Ollama) Contextual rationale Fallback handler

⊕ **Cross-Reference Expansion Layer**

Regex-based IS code detection in retrieved chunks → citation network traversal → indirectly related standards added to result set

04 CHUNKING & RETRIEVAL STRATEGY

CHUNKING DESIGN

- Each IS standard → paragraph-level chunks (not document-level)
- Rich metadata per chunk: chunk_id, is_code, title, section_id, section_name, page ref
- Structured sub-components: physical/chemical requirements, table captions & rows
- PDFs → semi-structured JSON knowledge base (dataset.json)
- Chunk overlap ensures clause boundaries don't split context

RETRIEVAL STRATEGY

- Intent Graph detects domain + IS code → structured intent object (section_id, content_type, confidence)
- O(1) metadata filter: only relevant section's FAISS entries compared
- all-MiniLM-L6-v2 embeds query into same 384-dim space as index
- FAISS cosine similarity → Top-K chunks (5–20)
- Cross-reference layer: regex extracts IS codes from results → adds cited standards

05 DEMO HIGHLIGHTS

From product description to BIS standard — live walkthrough

SAMPLE INPUT → Product Description

"We manufacture fly ash-based AAC blocks for load-bearing walls in residential construction."

#1 IS 1489	**Portland Pozzolana Cement** WHY: Fly ash content in AAC blocks requires PPC specification compliance for compressive strength.
#2 IS 2185	**Concrete Masonry Units** WHY: Load-bearing AAC blocks fall under this standard for block dimensions and structural properties.
#3 IS 6441	**Autoclaved Cellular Concrete** WHY: Direct standard for AAC block manufacturing, curing, and density classification.

Aura UI • Glassmorphic interface • IS code highlighting • Source citations per result

06 EVALUATION RESULTS

100%

Hit Rate @3

Target: >80%

0.883

MRR @5

Target: >0.7

0.03s

Avg. Latency

Target: <5s

WHAT DRIVES THESE NUMBERS

100% Hit@3 →	Metadata pre-filter eliminates cross-domain noise before semantic search runs
0.883 MRR →	Cross-encoder reranking resolves rank swaps between near-identical standards
0.03s latency →	FAISS index pre-built offline; query time = 1 embedding call + 1 nearest-neighbor search

166× below latency target • 26% above MRR target • 20% above Hit Rate target

07 IMPACT ON MSEs

Accelerating compliance for India's 63 million micro & small enterprises

BEFORE — Manual Process

- Weeks of manual document search
- Consultant fees for standard identification
- Risk of missing applicable standards
- Delayed product certification & market entry
- No plain-language explanation of requirements

AFTER — ARIVANTI

- Top 3–5 standards in seconds (0.03s retrieval)
- Plain English rationale for each recommendation
- 100% coverage of SP 21:2005 dataset
- No consultant required — self-service compliance
- Traceable citations for every recommendation

Every week saved in compliance = faster revenue for India's smallest manufacturers.

08 TEAM & ACKNOWLEDGEMENTS

TEAM

- Team Member 1 — RAG Architecture & Backend
- Team Member 2 — Intent Graph & Query Engine
- Team Member 3 — Data Ingestion & Evaluation

ACKNOWLEDGEMENTS

for SP 21:2005 dataset

Hackathon Organizers

for the problem statement
and evaluation framework

Open Source Community

TECH STACK

Python • **FastAPI** | FAISS • all-MiniLM-L6-v2 | **Llama 3** • **Ollama** | Aura UI (Glassmorphic) | **SP 21:2005 Dataset**